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PALAEOMAGNETISM OF SOME ROCKS FROM PENINSULAR INDIA AND KASHMIR

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ABSTRACT

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A palaeomagnetic study of seven sites in redbeds of the Late Precambrian Bhander and Rewa Series of the Upper Vindhyan System confirms that their original magnetization was extensively overprinted during the Early Tertiary, possibly related to the extrusion of the Deccan Traps about 65 Ma ago. Careful thermal demagnetization at temperatures close to the Curie Temperature of hematite revealed the primary magnetization in 100 of 121 specimens investigated. The resulting palaeomagnetic pole for the Upper Vindhyan System lies at 51.0S 37.8E. A combination with all previous results gives an overall palaeomagnetic pole at 47.3S 32.7E (N = 18, K = 35.5, A_{95} = 5.8°). Twelve samples from the Gwalior Traps (1830 Ma) give a palaeomagnetic pole at 16N 160.5E after magnetic cleaning.

Twelve flows collected from the Permo-Carboniferous Panjal Traps of Kashmir give mean direction D = 156.5, I = +32.5 (k = 19.8, α_{95} = 9.9°) with a positive fold test. The palaeomagnetic pole (32N 282E), however, lies close to that observed for Deccan Trap times in India. It appears that the magnetization of the Panjal Traps was acquired during the Early Tertiary Himalayan uplift following which they were tilted to their present attitudes.

INTRODUCTION

For many years palaeomagnetic studies of rocks from the Indian subcontinent have been concentrated on those formations whose intensity of magnetization could be measured with relatively simple and unsophisticated equipment. Considerable emphasis has therefore been placed on the study of various igneous rocks from the Mesozoic (Deutsch et al., 1958; Sahasrabudhe, 1963; Radhakrishnamurthy, 1963; McDougall and McElhinny, 1970; Klootwijk, 1971; Wensink and Klootwijk, 1971; Wensink, 1973; Verma and Mital, 1972, 1974; Kono et al., 1972) and from the Precambrian (Athavale et al., 1963, 1970; Verma et al., 1968; Klootwijk, 1974a, 1975a). In addition some studies were made of redbeds from the Precambrian

Vindhyan System (Sahasrabudhe and Mishra, 1966; Athavale et al., 1972) and from some of the Gondwana formations (Verma and Bhalla, 1968; Bhalla and Verma, 1969). More recently very careful studies of these redbeds have shown (Wensink, 1968, 1975; Klootwijk, 1973, 1974b, 1975b) that it is necessary to carry out detailed thermal demagnetization to high temperatures because of the presence of a persistent secondary magnetization of apparent Early Tertiary age. These authors have associated this overprint with the extrusion of the Deccan Traps extensively over the Peninsular Shield region. It is not at all clear how such widespread remagnetization of the sediments could result from the extrusion of thick lava piles except very close to the lava contact. Klootwijk (1975b, 1976) suggested the remagnetization resulted from the acquisition of a viscous partial thermoremanence due to sustained heating at temperatures between 100° and 200°C while the traps were being extruded over a prolonged period. This effect coupled with chemical remagnetization resulting from the associated hydrothermal effects is regarded as the source of the hard secondary magnetization. Whatever the cause, the presence of this hard secondary component in sediments from the Peninsular Shield has cast considerable doubt on the validity of some of the earlier studies.

Part of the problem has arisen because of the general lack of availability of sensitive magnetometers capable of measuring the very small primary component that is still preserved after thermal demagnetization close to the Curie Temperature. In this paper we report further results from the Bhander and Rewa Series of the Vindhyan System that confirms the extensive presence of a hard secondary component of apparent Early Tertiary age. We also report the first palaeomagnetic measurements from the Permo-Carboniferous Panjal Traps of Kashmir. Until this time the rather curious fact that these volcanics are extremely weakly magnetized has prevented investigators in India from carrying out any palaeomagnetic studies of them (R.K. Verma and R.N. Athavale, personal communication).

SAMPLING AND MEASUREMENTS

The samples investigated in this study were collected during an expedition to India in 1967 carried out with the assistance of the Geological Survey of India. The sampling localities are shown in Fig. 1. Seven sites were collected from the Upper part of the Vindhyan System, six sites (eighteen samples) from quarries in the vicinity of Fatehpursikri 50 kilometres or more to the west of Agra. These sites are all in the Bhander Series. A single site (three samples) was collected from the Rewa Series further to the east in the vicinity of Mirzapur. Previously, samples of the Bhander Series have been investigated by Athavale et al. (1972) and Klootwijk (1973). Athavale et al. (1972) also reported measurements from one site in the Rewa Series. Measurements on the underlying Kaimur Series have previously been reported by Sahasrabudhe and Mishra (1966).

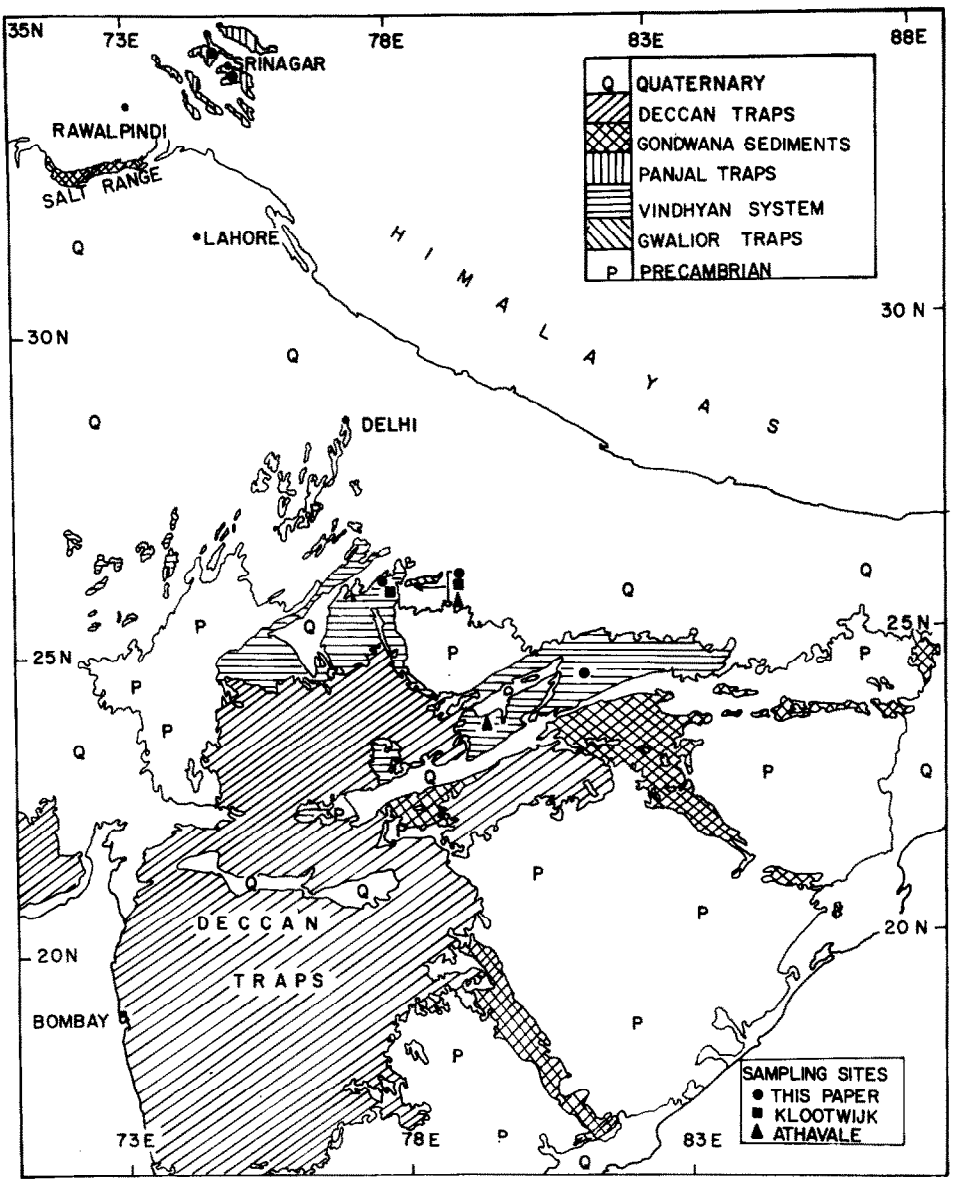


Fig. 1. Generalised geological map of part of the Indian sub-continent showing sampling localities of rocks investigated in this study and by Athavale et al. (1963, 1972) and Klootwijk (1973, 1974a).

The age of the Vindhyan System has always been the subject of much debate. Isotopic dating has been carried out by Tugarinov et al. (1965) using K—Ar measurements on glauconites and by Crawford and Compston (1970) who dated a kimberlite pipe intruding the Kaimur Series using the Rb—Sr

method. The Rb—Sr age of 1140 ± 12 Ma is rather older than that of about 950 Ma obtained from the glauconites, indicating that the Kaimur Series is at least 1000 Ma old and possibly could be as old as 1200 Ma. Possible correlation of the Bhander and Rewa Series with the Jodhpur Sandstones of Rajastan suggests these rocks could extend up into the lower Palaezoic. The Jodhpur Sandstones overlie the Malani Rhyolites dated by Crawford and Compston (1970) at 745 ± 10 Ma using the Rb—Sr method.

Twelve samples were collected from the Gwalior Traps in the vicinity of the Gwalior Fort. Material from these samples was used by Crawford and Compston (1970) to obtain an age of 1830 ± 200 Ma using the Rb—Sr method. Samples from these localities have previously been investigated by Athavale et al. (1963) and Klootwijk (1974b). Athavale et al. (1963) incorrectly assigned these rocks to the Bijawar Series and, as Crawford and Compston (1970) have shown, the correlation of the Bijawar and Gwalior Traps is incorrect. The Bijawar Traps are about 600 Ma older than the Gwalior Traps.

Thirteen flows (38 samples) were collected from the Panjal Volcanic Traps in the vicinity of Srinagar in Kashmir. Following Krishnan (1968) the late Palaeozoic succession in Kashmir is as follows:

Zewan beds with *Protoretepora Ampla* bed near the base	Middle and Upper Permian
Gangamopteris beds	Permo-Carboniferous
Panjal Traps	Permo-Carboniferous
Agglomeratic Slates	Upper Carboniferous
Fenestella shales	Middle Carboniferous
Syringotheris limestone	Lower Carboniferous
Muth Quartzite	Devonian

These flows were collected from two quarries near Srinagar town (Atwai quarries), six flows in a succession up Shankar Acharya Hill where the traps unconformably overlie Upper Carboniferous (*Eurydesma*) beds and three flows from the neighbouring Hari Parbat Hill. A single flow was collected in a road cutting in the Wular Lake region where the traps are directly overlain by Triassic limestone.

The samples were originally measured on a PAR SM1 Spinner Magnetometer but the surprisingly weak intensities observed in the Panjal Traps necessitated putting these aside until they could be processed using a Digico Complete Results Magnetometer. Although some of the initial measurements on all the collections were made on the SM1 Magnetometer the final analyses and cleaned measurements were made on the Digico Magnetometer. Alternating field demagnetization and thermal demagnetization procedures were carried out on apparatus described by McElhinny (1966) and McElhinny et al. (1970), respectively.

VINDHYAN ROCKS

Thermal demagnetization tests on specimens of these redbeds showed the presence of up to three components of magnetization. In general there was a

viscous component directed along the present earth's field which was removed in temperatures up to 500°C. However, many specimens required demagnetization in temperatures between 600 and 665°C to remove another component of magnetization directed along the same direction of magnetization as observed in the Early Tertiary normally magnetized Deccan Traps. The response of specimens from the same sample was often different, sometimes this hard secondary component could be cleaned before reaching the Curie Temperature of hematite and sometimes not. As a result of this observation six specimens from each sample (except at site 3 where lack of material prevented this) were processed and were all carefully demagnetized at successive temperatures between 550 and 670°C. The varying types of thermal demagnetization curves obtained from specimens with one, two or three components of magnetization are illustrated in Fig. 2.

Only at sites 4 and 5 (Table I) could all specimens be cleaned of the hard secondary magnetization. At the other sites up to one third of the specimens retained the hard secondary magnetization at 670°C. Of the 131 specimens investigated twenty-one retained the hard secondary magnetization as illustrated in Fig. 3. The mean direction is close to that found for the Early Tertiary normally magnetized Deccan Traps by Wensink (1973) (see Table I).

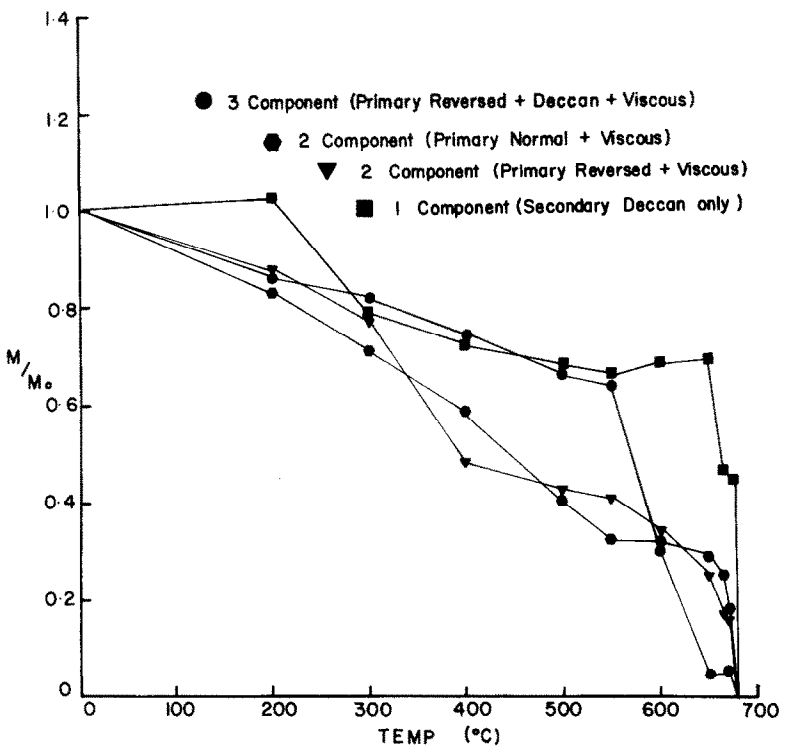


Fig. 2. Thermal demagnetization curves of samples from the Upper Vindhyan System.

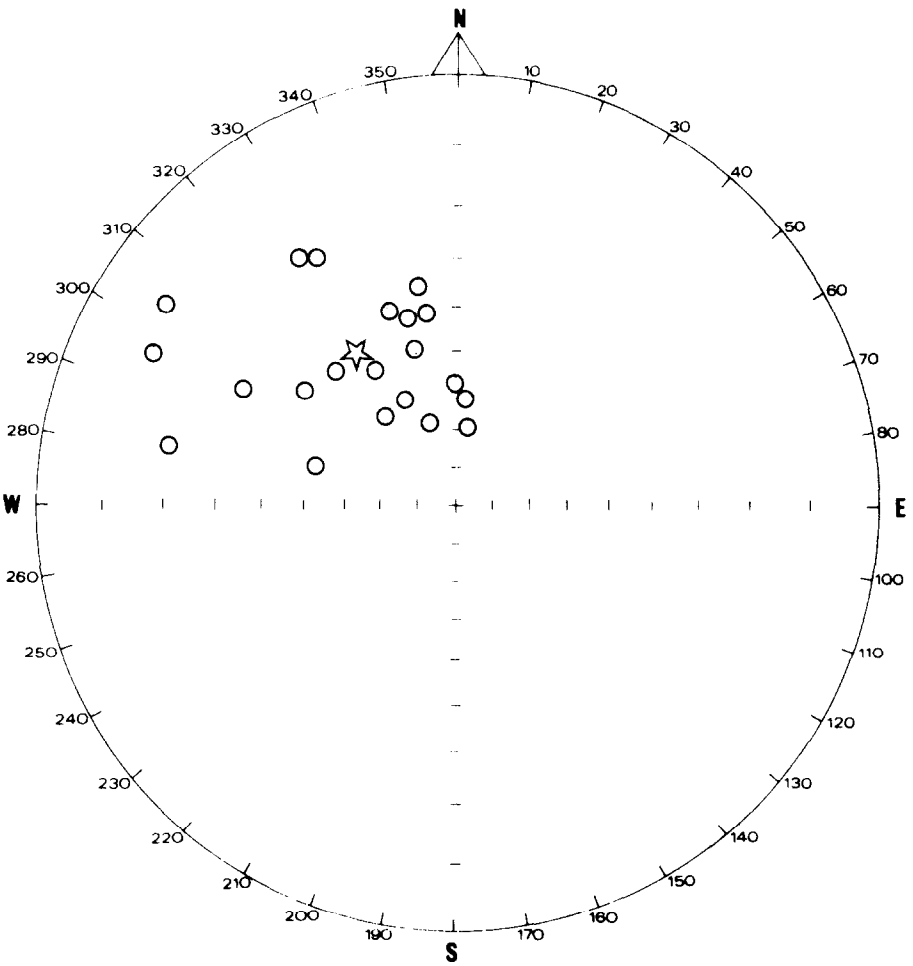


Fig. 3. Overprinted magnetization acquired during Early Tertiary times that could not be cleaned from twenty one specimens of Upper Vindhyan sandstones. Open circles on the upper hemisphere; star is the mean direction.

In calculating the overall mean for each site (Table I) these twenty one specimens were omitted. One of the seven sites has normally magnetization and this appears to be the only normally magnetized site so far observed in the Upper Vindhyan System. The remaining seventeen sites studied here and by Athavale et al. (1972) and Klootwijk (1973) are reversely magnetized and form a coherent group around 210, +10 (Fig. 4).

Figure 4 shows that there appears to be no distinction between the two sites so far studied from the Rewa Series and those from the Bhander Series. When the data from all studies are combined together (Table II) the best estimate of the palaeomagnetic pole for Upper Vindhyan times is given by combining all eighteen site poles. As already noted by Athavale et al. (1972)

TABLE I

Directions of magnetization observed in Upper Vindhyan redbeds after thermal cleaning

Site	Unit	Locality	$N(n)$	D	I	R	Pole position
1	Rewa Series	24.5N 81.0E	12(6)	221.6	−8.0	9.3635	45.0S 11.3E
2	Bhander Series	27.0N 77.5E	13(5)	182.5	+4.3	9.2931	60.7S 72.2E
3	Bhandee Series	27.0N 77.5E	11(2)	35.4	−36.2	10.5409	31.6S 37.6E
4	Bhander Series	27.0N 77.5E	18(0)	205.1	+18.0	16.8736	46.3S 40.0E
5	Bhander Series	27.0N 77.5E	18(0)	196.0	+9.8	16.0632	54.4S 49.2E
6	Bhander Series	27.0N 77.5E	12(6)	203.2	−3.1	9.0172	56.1S 32.3E
7	Bhander Series	27.0N 77.5E	16(2)	203.2	−0.6	11.9347	55.2S 33.7E
	overall result	7 sites		203.4	+8.1	6.6574	51.0S 37.8E
	Early Tertiary overprint	(21)		323.9	−44.4	19.1212	26.5N 293.6E

N = no. of cleaned specimens used in calculation; n = no. of specimens that could not be cleaned of the hard secondary magnetization; D = declination east of true north; I = inclination position downwards; R = resultant of N unit vectors.

this mean pole position lies close to those found by McElhinny (1970) and Wensink (1972) for Cambrian sediments of the Salt Range. This supports the view that they should be correlated.

GWALIOR TRAPS

Alternating field demagnetization was carried out on four specimens from each of the twelve samples. The demagnetization curves (Fig. 5) are very similar and show that about 90% of the N.R.M. is removed by 20 mT peak field at which point the curves flatten out. The remaining 10% consists of magnetization capable of withstanding high magnetic fields. Klootwijk

TABLE II

Summary of palaeomagnetic data for the Upper Vindhyan System

Investigator	Unit	N	K	A_{95}	Palaeomagnetic pole
Athavale et al. (1972)	Rewa	1	—	—	35.7S 41.2E
Athavale et al. (1972)	Bhander	3	371	6.4	31.8S 19.0E
Klootwijk (1973)	Bhander	7	140	5.1	51.4S 34.2E
this paper	Rewa	1	—	—	45.0S 11.3E
this paper	Bhander	6	36.5	11.2	51.3S 42.7E
overall mean	Rewa	2	—	—	41.3S 27.3E
overall mean	Bhander	16	36.2	6.2	48.1S 33.6E
overall mean	U. Vindhyan	18	35.5	5.8	47.3S 32.7E

N = no. of sites; K = precision parameter; A_{95} = radius of circle of 95 percent confidence about mean.

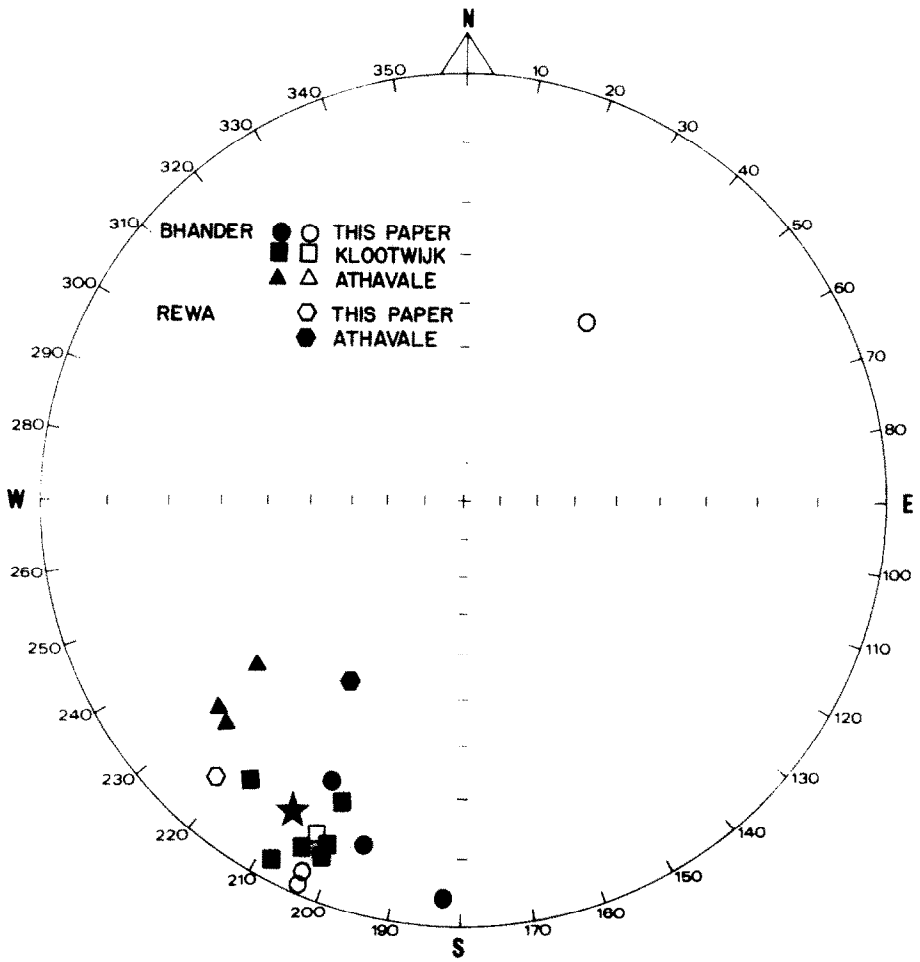


Fig. 4. Mean directions after cleaning observed in eighteen sites studied in the Upper Vindhyan sandstones in this study and by Athavale et al. (1972) and Klootwijk (1973). Open symbols on the upper hemisphere, solid symbols on the lower hemisphere.

(1974a) has previously noted that there appear to be "less-hard" (after demagnetization at 20 mT) and "hard" (after demagnetization at 80 mT) components of magnetization present in these rocks. The hard component has the same declination but slightly steeper inclination than the less-hard component. To investigate this possibility the 48 specimens were first demagnetized in 20 mT and then again at 80 mT (Fig. 6, Table III). The scatter increased slightly at 80 mT and the inclination bec ame steeper as observed by Klootwijk (1974a). However to check that the scatter and change of inclination was not due to imperfect demagnetization, the specimens were reversed in the demagnetization apparatus and again demagnetized at 80 mT. The single and double demagnetization results show the

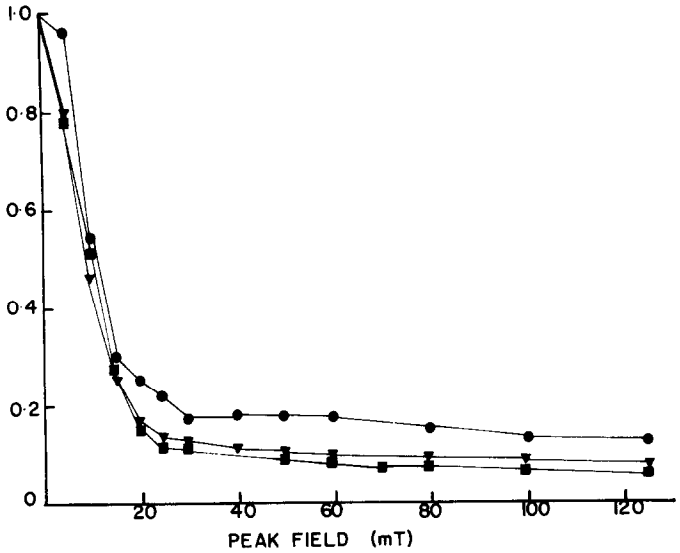


Fig. 5. Alternating field demagnetization curves of three samples of Gwalior traps.

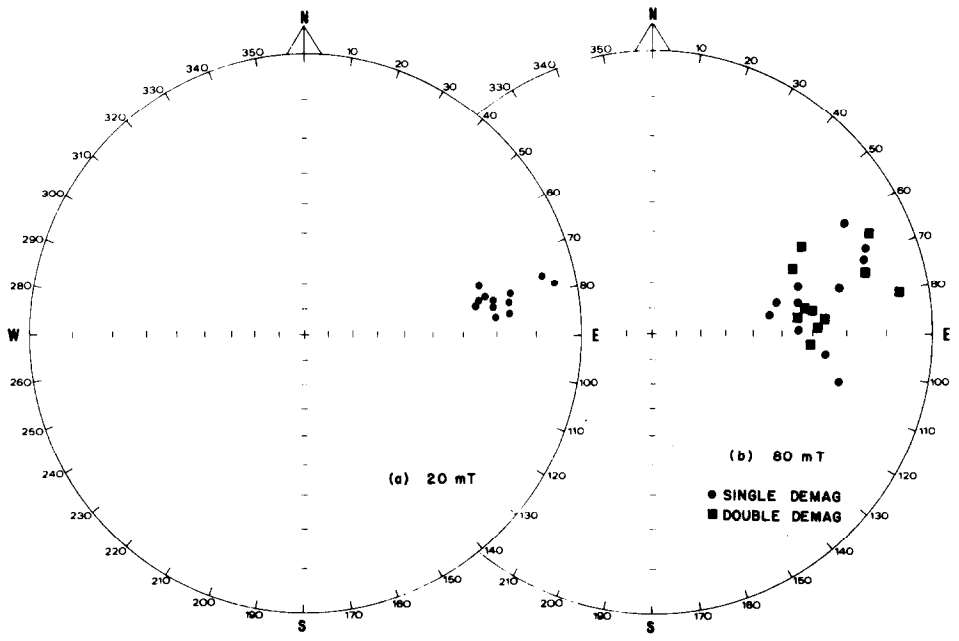


Fig. 6. Directions of magnetization observed in twelve samples of Gwalior traps (a) after cleaning at 20 mT (b) after cleaning at 80 mT using single demagnetization (circles) or double demagnetization (squares).

TABLE III

Directions of magnetization observed in studies of the Gwalior Traps

Investigators	N	D	I	k	α_{95}	Pole position
Athavale et al. (1963)	7	70	+3	—	18	19N 176E
Klootwijk (1974a) (less-hard component)	15 *	79.5	+8	56	12.5	11N 169E
Klootwijk (1974a) (hard component)	23 *	78	+34.5	369	5	19N 155.5E
this paper (20 mT)	12	79.7	+17.7	126.4	3.8	13N 164E
this paper (80 mT)	12	78.5	+25.8	47.5	6.3	16N 160.5E

Site location 26N 78E. Symbols as in Table I.
* N = 4 sites.

same mean direction and similar scatter (Fig. 6b). There is no doubt therefore that the 80 mT results show a significantly steeper inclination (by 8.1°) than that at 20 mT (Table III, Fig. 6).

The results presented here concur with the findings of Klootwijk (1974a) and suggest that the samples studied by Athavale et al. (1963) were not subjected to high enough alternating fields. A similar situation has recently been revealed by Schmidt (1976) with respect to Mesozoic igneous rocks studied more than a decade ago in eastern Australia. The results determined by us and by Klootwijk (1974a) at 80 mT are therefore most probably representative of the primary magnetization acquired when these traps were extruded.

PANJAL TRAPS

Four specimens were examined from each of the samples collected. Natural remanent magnetization intensities ranged from 0.1 to about 20 mA m^{-1}. The stability of samples to both alternating field and thermal demagnetization varied considerably; extreme examples are shown in Fig. 7. Sample PT2 showed exceptional stability with a single thermally discrete component with blocking temperature at the Curie Point of magnetite. On the other hand sample PT25 contained a very soft component in the direction of the present earth's field removed in 10 mT or 300°C. Thereafter the sample was stable up to the Curie Point of magnetite. All samples cleaned in alternating fields between 10 and 30 mT and were usually stable and persistent under thermal demagnetization up to the Curie Point of magnetite. No high temperature component could be distinguished that differed from the a.f. cleaned directions.

Twelve of the thirteen flows gave consistent directions upon cleaning (Table IV, Fig. 8). All were reversely magnetized and upon correcting for the dip of the flows the precision of the twelve flow directions increases from 9.6 to 19.8. For N = 12 this increase in precision is significant at the 95%

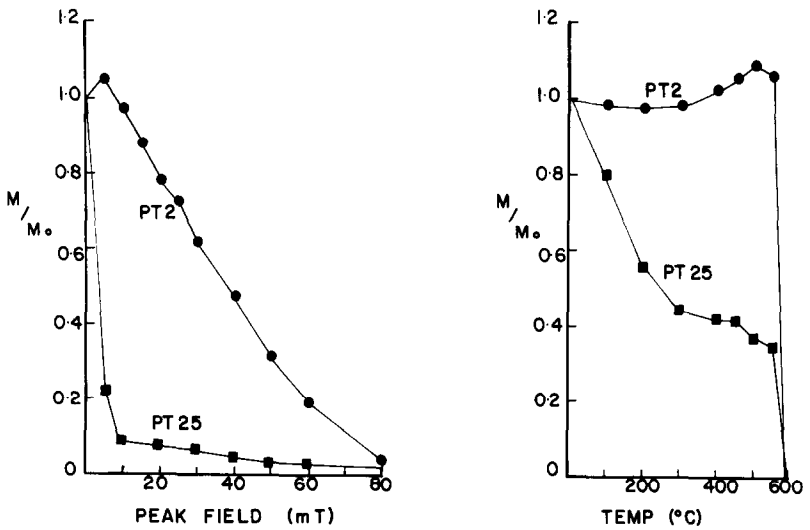


Fig. 7. Alternating field and thermal demagnetization curves of two samples of the Panjal traps of Kashmir.

TABLE IV

Directions of magnetization observed in the Panjal traps of Kashmir after alternating field cleaning

Flow	N	In situ		Bedding	
		D_u	I_u	D_c	I_c
1	3	251	+69	187	+35
2	3	33	+82	152	+50
3	3	144	+14	138	+52
4	3	148	+34	148	+34
5	3	147	+18	147	+18
6	3	167	+13	167	+13
7	2	146	+29	146	+29
8	3	161	+16	161	+16
9	3	148	+45	148	+45
10	3	166	+39	168	+24
11	3	165	+38	167	+23
12	2	120	+57	136	+48
mean	12	154 (k = 9.6,	+40 α_{95} = 16.8)	156.5 (k = 19.8,	+32.5 α_{95} = 9.9)

Symbols as in Table I. Mean locality 34.1N 74.8E.

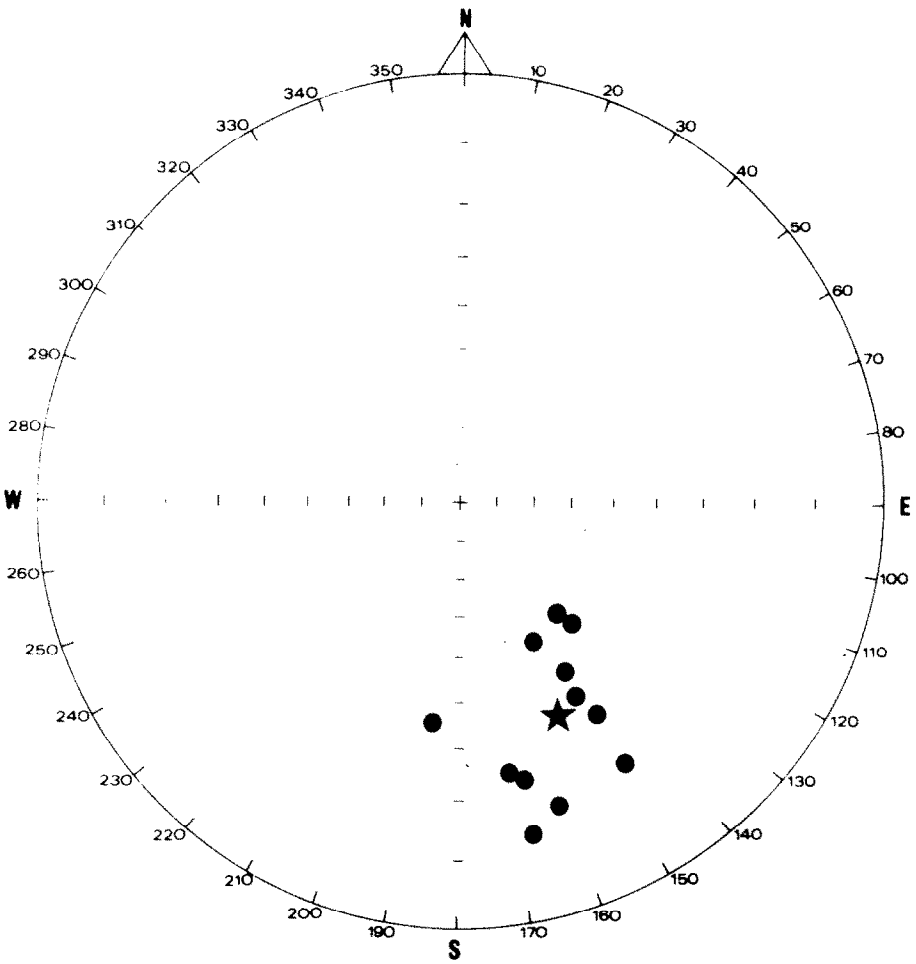


Fig. 8. Directions of magnetization observed in twelve flows of Panjal traps after magnetic cleaning. Symbols as in Fig. 4; star is mean direction.

confidence level (McElhinny, 1964). This would appear to suggest that the magnetization was acquired in pre-Tertiary times before the Himalayan uplift. However, the mean pole position for these rocks lies at 32N 282E and this is almost identical with that observed in the Deccan Traps (Wensink, 1973). It is far removed from that found in Permian rocks from the Salt Range (Wensink, 1975) and from the Peninsular Shield (Wensink, 1968; Klootwijk, 1975b). However, there have been several episodes of folding in the Kashmir Himalayas during the Tertiary so that the positive fold test only serves to indicate that the magnetization of the Panjal traps was acquired during the Early Tertiary or beforehand. It appears therefore that these rocks may have acquired their present magnetization during the Early Terti-

ary Himalayan uplift before they were tilted to their present attitude. Further sampling of rocks in this region for palaeomagnetic studies will indicate whether or not this explanation is correct.

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